

Part I: Theory Questions

(Chapter-wise Collection of All Theory Concepts)

Module 1: Introduction to Programming

1. Define: algorithm, flowchart, compiler. **(W-24, W-18)**
2. Draw block diagram of computer and explain functionalities of various components of computer. **(S-25, S-19)**
3. Describe various symbols used for preparing flow chart. **(W-24, W-18, S-20)**
4. Brief about computer hardware and software. **(S-25)**
5. Explain high level language, machine level language and assembly level language. **(S-20, W-22)**
6. Differentiate between compiler and interpreter. **(S-20, W-21)**
7. List out various input and output devices. **(W-21)**

Module 2: Fundamentals of C Programming

1. Explain different type of operators used in C language with their precedence and associativity. **(W-24, W-18, W-22)**
2. Explain different bit-wise operators with the example. **(S-25, W-21, S-22)**
3. Explain getch(), getchar(), gets(), puts(). **(W-24, W-22, S-23)**
4. Describe local and global variable with example. **(W-18, W-21)**
5. Explain type casting (Implicit and Explicit) with example. **(W-18, S-23)**
6. What is constant? Explain different types of constants. **(S-22)**
7. Define data types in C. Explain primitive data types. **(W-21)**
8. Explain the structure of a C program. **(W-19)**
9. Demonstrate the use of conditional operator statement with an example. **(W-24)**
10. Explain formatted output using printf() statement. **(S-23)**
11. What do you mean by type conversion? Give example. **(W-18)**

Module 3: Control Structures

1. Explain the structure of switch case statement with example. (W-24, W-18, W-22, S-22)
2. Differentiate: Exit controlled Loop and Entry controlled Loop. (W-24, S-20)
3. Explain: a) break statement b) continue statement. (W-24, W-18, S-19, S-23)
4. Explain the structure of for loop with proper example. (S-25, W-18)
5. Explain while loop and do-while loop with example. (S-25, W-18, W-21)
6. Explain nested if-else statement with suitable example. (S-25, W-18)
7. Explain else-if ladder with example. (S-23)
8. Explain various looping control structures with suitable example. (S-19)
9. Define general form of 1) do while loop 2) Nested if 3) goto. (W-18)

Module 4: Arrays and Strings

1. Summarize the methods for initialization of One-Dimensional array. (W-24)
2. Explain one dimensional array, two-dimensional array and multi-dimensional array with their syntax and example. (S-25, S-19, W-21)
3. Explain following string manipulation function: strcat(), strcpy(), strcmp() and strlen(). (W-24, S-19, S-23, W-21)
4. Explain Declaration, initialization and Printing on screen of a String using suitable program. (S-25)
5. Why null value is used in string? Justify your answer with example. (W-18)
6. Describe array with example. Why it is necessary to give the size of an array in array declaration? (S-19, W-18)
7. What is a string? Explain with example to find a character from string. (W-18)

Module 5: Functions

1. What is UDF? Describe advantages of UDF. (W-18, W-21)
2. Explain recursion with example. List the advantages of recursion. (S-25, W-18, S-22)
3. Discuss parameter passing technique used in C with example (Call by value vs Call by reference). (W-18, S-23, S-25)
4. Classification of User defined function's components (elements). (W-18)
5. What is function? Explain function definition, function declaration and function call. (S-23)

Module 6: Pointers

1. What is pointer? Which arithmetic operations are valid on pointers? (S-25, S-19)
2. What is pointer to pointer? Write suitable example to demonstrate the concept. (S-25, S-20)
3. Explain with example to print the address of variable using pointer. (W-18)
4. Explain array of pointers with suitable example. (S-19)
5. Write True/False: Pointer is a variable which holds address of location where value of other variable is stored. (W-24)
6. Differentiate between array and pointer. (W-22)

Module 7: Structures and Unions

1. What is structure? Explain with example how to declare a structure and how to initialize it. (W-24, W-18, S-20)
2. Differentiate between structure and union. (W-24, S-25, W-21, S-22)
3. Define Union in 'C' with suitable example. (S-25, S-23)
4. Explain nested structure with example. (S-25, S-23)
5. What is structure? How to access the elements of structure? How to calculate size of structure? (W-18)

Module 8: File Handling

1. Explain various file management function in C. (S-25, W-21)
2. Explain fopen() and its mode with example. (W-24, W-18, S-22)
3. Explain error handling in file system with example. (S-25)
4. Explain the following File Handling functions: fseek(), ftell(), fread(), fwrite(), fscanf(), fprintf(), rewind(). (S-25, S-23)
5. Briefly explain any two file handling functions with an example. (W-24)
6. What is file? List out various file opening modes. (W-22)
7. Explain fopen() and its mode with example to write a string into file. (W-18)

Module 9: Dynamic Memory Allocation

1. Show the use of Malloc, Calloc, Free and Realloc in context of Dynamic memory allocation. (S-25, S-22, W-21)
2. Demonstrate the use of void * with example. (S-23)
3. Briefly explain memory management functions. (S-23)

Part II: Programming Problems

(Chapter-wise Collection of All Coding Questions)

Module 1: Introduction to Programming (Algorithms/Flowcharts)

1. Write an algorithm to check whether the entered number is Even or Odd. **(W-24)**
2. Write algorithm or flowchart for finding maximum out of three numbers entered by user. **(S-25, S-20)**
3. Write flowchart or algorithm to find area of a triangle. **(S-19)**
4. Draw flowchart to find factorial number. **(W-18)**
5. Draw the flow chart to find the sum of 1 to n numbers. **(S-20)**

Module 2 & 3: Fundamentals & Control Structures

1. Write a C program to find factorial of a number. **(W-24)**
2. Construct a C program to swap the values of two integers with use of only two variables. **(W-24, W-22)**
3. Write a program to print the triangle shown below. **(W-24)**
1
10
101
1010
10101
4. Write a program to check whether entered character is vowel or not? **(W-24, S-19)**
5. Write a C program to find the largest number amongst three numbers entered by user. **(S-25)**
6. Write a C program to check if entered integer is a prime number or not. **(S-25, W-18)**
7. Write a program to reverse a given number. **(S-19, W-22)**
8. Write a program to find $1 + 1/2 + 1/3 + 1/4 + \dots + 1/n$. **(S-19)**
9. Write a program to print all Armstrong numbers in a given range. **(S-19, S-20)**
10. Write a C program to input an integer number and check last digit of number is even or odd. **(W-18)**
11. Write a program to print Fibonacci series up to N numbers. **(S-20, S-22)**
12. Write a program to check whether the given number is Palindrome or not. **(W-21, S-23)**

13. Write a program to find out the maximum number from integer array of 5 elements. (S-23)

Module 4: Arrays and Strings

1. Write a program to add all the elements of an array. (W-24)
2. Construct a C program to add 3X3 matrix. (W-24, S-22)
3. Write a program to display transpose of given 3*3 matrix. (S-19)
4. Write a program to multiply two 3x3 matrices. (S-20)
5. Write a C program to sort the array elements in ascending order. (S-22, W-21)
6. Write a C program to count total number of vowel in given string. (S-23)
7. Write a program to find out the maximum number from integer array of 5 elements. (S-23)
8. Write a C program to check whether the given string is palindrome or not. (S-22)
9. Write a C program to find the length of a string without using library function. (W-21)

Module 5: Functions

1. Create a function to check number is prime or not. (W-18)
2. Create a function to swap the values of two variables. (W-18)
3. Write a program to calculate nCr using user defined function. (S-19)
4. Write a C program to find factorial of a number using recursion. (S-20, W-21)
5. Develop a program in C to check the entered number is prime or not by creating a user-defined function named check_prime(). (W-24)

Module 7: Structures

1. Define a Structures which contains details of a Cricketer: Name of Player, Team name, Total run scored, Batting average. (W-24)
2. Define a structure data type called time_struct containing three members: integer hours, minutes, second. Develop a program that would assign values to individual member and display the time in following format: HH:MM:SS (W-18)
3. Define a structure called "student" with members: name, roll no, and marks. Write a program to read and display details of 5 students. (S-20, W-22)
4. Write a program to demonstrate the use of nested structure. (S-22)

Module 8: File Handling

1. Write a C program to copy content of one file to other with the help of file handling functions. (W-24, S-20, W-21)
2. Write a program to read data from a file and display it on the screen. (S-22)
3. Demonstrate the use of fopen() and fprintf() function of file handling. (S-23)
4. Write a program to write a string into a file using fopen(). (W-18)
5. Write a C program to append content to a file. (W-22)
6. Write a C program to count number of characters, spaces, tabs and newlines in a file. (S-20)



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